

NAME

aprecenter -- Recenter apertures automatically

USAGE

aprecenter input

PARAMETERS**input**

List of input images in which apertures are to be recentered.

apertures = ""

Apertures to recenter, resize, trace, and extract. This only applies to apertures read from the input or reference database. Any new apertures defined with the automatic finding algorithm or interactively are always selected. The syntax is a list comma separated ranges where a range can be a single aperture number, a hyphen separated range of aperture numbers, or a range with a `*` for example, "1, 3-5, 9-12x2".

references = ""

List of reference images to be used to define apertures for the input images. When a reference image is given it supersedes apertures previously defined for the input image. The list may be null, "", or any number of images less than or equal to the list of input images. There are three special words which may be used in place of an image name. The word "last" refers to the last set of apertures written to the database. The word "OLD" requires that an entry exist and the word "NEW" requires that the entry not exist for each input image.

interactive= no

Run this task interactively? If the task is not run interactively then all user queries are suppressed and interactive aperture editing is disabled.

find = yes

Find the spectra and define apertures automatically? In order for spectra to be found automatically there must be no apertures for the input image or reference image defined in the database.

recenter= yes

Recenter the apertures?

resize = no

Resize the apertures?

edit = yes

Edit the apertures? The *interactive* parameter must also be yes.

line = INDEF

The dispersion line (line or column perpendicular to the dispersion axis) to be used in recentering the spectra. A value of INDEF selects the middle of the image.

nsum = 1

Number of dispersion lines to be summed or medianed. The lines are taken around the specified dispersion line. A positive value takes a sum and a negative values selects a median.

aprecenter = " "

List of apertures to be used in shift calculation.

npeaks= INDEF

Select the specified number of apertures with the highest peak values to be recentered. If the number is INDEF all apertures will be selected. If the value is less than 1 then the value is interpreted as a fraction of total number of apertures.

shift = yes

Use the median shift from recentering the selected apertures to apply to all apertures. The recentering is then a constant shift for all apertures. The median is the average of the two central values for an even number of apertures.

ADDITIONAL PARAMETERS

I/O parameters and the default dispersion axis are taken from the package parameters, the default aperture parameters are taken from the task **apdefault**, automatic aperture finding parameters are taken from **apfind**, and parameters used for centering and editing the apertures are taken from **apedit**.

When this operation is performed from the task **apall** all parameters except the package parameters are included in that task.

DESCRIPTION

For each image in the input image list, the aperture center positions are redefined by centering at the specified dispersion line using the **center1d** algorithm with centering parameters from **apedit**. Normally this is done when inheriting apertures from an aperture reference image. The recentering does not change the "trace" of the aperture but simply adds a shift across the dispersion axis.

There are several recentering options. Each selected aperture may be recentered independently. However, if some or all of the spectra are relatively weak this may actually be worse than using the reference apertures defined by strong spectra or flat fields in the case of fibers or aperture masks. One may select a subset of apertures used in calculating shift. This is done with the *aprecenter* list of aperture numbers (see **ranges** for the syntax) and/or by selecting a specific number or fraction of the apertures with the strongest peak values. The list selection is done first and the strongest remaining apertures are used to satisfy the **npeaks** value. Though some or all of the apertures may be recentered independently the most common case of recentering reference apertures is to account for detector shifts. In this case one expects that any shift should be common to all apertures. The *shift* parameter allows using the new centers for all selected apertures to compute a median shift to be added to ALL apertures. Using a median shift for all apertures is the default.

The *find* parameter allows automatically finding apertures if they are defined for the image or by a reference image. Since the purpose of this task is to recenter reference apertures it is usually the case that reference images are used and apertures are not defined by this task. One case in which the apertures from the image itself might be recentered is if one wants to use a different dispersion line. The *resize* parameter may be used to adjust the widths in a variety of ways based on the spectra profiles specific to each image. The aperture positions and any other parameters may also be edited with the aperture editing function if selected by the *apedit* parameter and the task is run interactively. The recentering algorithm may be run from the aperture editor using the 'g' keystroke.

If the task is interactive the user is queried whether to perform various steps on each image. The queries may be answered with one of the four values "yes", "no", "YES" and "NO", where an upper case response suppresses all further queries to this question.

The aperture recentering algorithm may be selected from nearly every task in the package.

EXAMPLES

```
cl> aprecenter newimage reference=flat
```

REVISIONS**APRECENTERV2.11**

The "apertures" parameter can be used to select apertures for resizing, recentering, tracing, and extraction. This parameter name was previously used for selecting apertures in the recentering algorithm. The new parameter name for this is now "aprecenter".

SEE ALSO

center1d, ranges, apfind, apresize, apedit, apall